Designing a Structured Onboarding Framework to Improve Time-to-Effectiveness for Security Engineering Teams

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***Abstract*— Security Engineering teams are on the front lines of the fight against adversaries attempting to compromise the systems and data of organizations. Despite this role being complex and high-risk, onboarding is often informal, inconsistent, and reliant on ad hoc knowledge transfer. This lack of structure contributes to increased time-to-effectiveness and security risk. This paper addresses the lack of a structured onboarding approach tailored to Security Engineering teams. Using a design-oriented, mixed-method approach, stakeholder input from managers, experienced engineers, and recent joiners was analyzed. From this analysis, themes were developed that encompass the feedback and paint a clear picture of the difficulties encountered in typical onboarding processes. These themes were used to guide the design of a formal onboarding framework that provides recommendations and artifacts to assist organizations in developing customized onboarding guides that cover their specific workflows and roles emphasizing role clarity, operational readiness, and onboarding efficiency. Evaluation of stakeholder feedback suggests they agree that this framework improves these factors while remaining adaptable across organizations. The conclusion drawn from this evaluation is that organizations would improve time-to-effectiveness by implementing this onboarding framework.**

I. Introduction

Security Engineering teams are responsible for designing, reviewing, and implementing controls that protect critical systems and data across complex environments, such as cloud infrastructure, identity platforms, and custom applications. Misconfigurations, delayed access, or unclear decision-making can have immediate and high-impact security consequences. As organizations scale to meet increasing security demands, new engineers are onboarded with diverse backgrounds and varying familiarity with internal architectures, workflows, and threat models. Effective onboarding is therefore essential to maintaining security effectiveness and operational continuity.

Despite this importance, onboarding within Security Engineering teams is often informal, inconsistent, and dependent on tribal knowledge. While corporate orientation and access provisioning are often well established, security-specific onboarding – including system architecture, decision-making expectations, and risk ownership – is typically ad hoc. New hires are expected to learn through scattered documentation and peer assistance, placing an added burden on senior engineers and delaying meaningful contribution.

I was motivated to pursue this project because improving onboarding effectiveness for Security Engineering teams offers a high-impact way to strengthen security outcomes without requiring new technology or tooling. My interest was shaped by firsthand experience navigating an unstructured onboarding process, which required extensive self-directed effort and reliance on informal mentorship to understand security architecture, workflows, and expectations. Drawing on my background in process improvement, I approached onboarding as an operational system that directly influences early effectiveness. This project aligns with the goals of the Cybersecurity Practicum by applying cybersecurity theory, risk management principles, and organizational process design to produce practical, reusable artifacts that address a real-world challenge faced by Security Engineering teams.

The core problem addressed by this project is the absence of a structured, repeatable onboarding framework tailored specifically to Security Engineering teams that measurably improves time-to-effectiveness. As I experienced, onboarding outcomes without structure can depend heavily on individual initiative or the availability of informal mentorship. Without an intentional, scalable process aligned with security responsibilities, operational and security risk is introduced by extended ramp-up periods, reduced team velocity, and increased likelihood of misconfigurations, incomplete threat modeling, or inconsistent security decisions. Over time, reliance on informal knowledge transfer also increases key-person risk, contributes to burnout, and negatively affects morale, retention, and overall security posture.

The rest of this paper is organized as follows. First, I begin by establishing the background and related work that shaped the problem space and revealed the gaps driving this project. Second, I outline the methodology I used to collect and analyze the data. Following this, I do a walkthrough of the key findings that emerged. Next, I propose the solution directly guided by those findings. Then, I evaluate that solution to understand where it performs well and where it falls short. From there, I discuss what the results mean in a broader context. In conclusion, I come full circle to tie the problem and solution together. Finally, I close by outlining where the work can go next. I will begin by discussing the research and prior work that acted as background for my project in the next section.

II. Background & Related Work

Security Engineering onboarding sits at the intersection of cybersecurity governance, workforce development, and organizational socialization. While numerous industry standards define the competencies, controls, and responsibilities expected

of security personnel, few provide guidance on how organizations should structure onboarding to achieve those expectations, and none do so in a systematic fashion. As a result, many Security Engineering teams rely on informal, inconsistent, or ad hoc onboarding practices, a pattern reflected by personal experience and in stakeholder feedback gathered for this project.

A. Industry Standards

Several widely adopted cybersecurity standards establish expectations for the knowledge, skills, and responsibilities of security personnel. The NIST NICE Workforce Framework for Cybersecurity provides a comprehensive taxonomy of tasks, knowledge, and skills required across cybersecurity roles. As the document notes, the framework "is a reference resource for those seeking to describe the cybersecurity work their organization does, the people who will carry out the work, and the onboarding learning that will be needed to do that work effectively [1]." While NICE is valuable for defining what a security engineer must ultimately be capable of, it does not prescribe how organizations should structure onboarding to achieve those competencies.

Similarly, NIST 800-53 establishes requirements for training, access control, and auditability, emphasizing that personnel with security responsibilities must receive specialized training [2]. This requirement materially differentiates the onboarding for security engineering from onboarding of fields like software engineering, where privileged access, risk exposure, and regulatory obligations are far less pronounced.

These controls imply a structured, risk-aligned onboarding process; however, like NIST NICE [1], NIST 800-53 [2] stops short of defining a holistic onboarding process that integrates access provisioning, documentation, mentorship, and operational readiness. Together these standards establish what security engineers must know and why onboarding matters, but they do not offer a unified model for how onboarding should be executed.

B. Organizational Onboarding Models

General organizational onboarding research emphasizes that new hire time-to-effectiveness is strongly influenced by early role clarity, structured social integration, and timely exposure to job-relevant tasks. Foundational socialization theory and subsequent empirical studies show that reducing ambiguity, providing clear expectations, and enabling guided early task engagement accelerate adjustment and improve early performance outcomes [3-5]. These models highlight onboarding as a deliberate developmental process rather than a purely administrative one, underscoring the importance of structured learning pathways, consistent communication, and intentional support mechanisms during the first months of employment. Since they were scoped in a general manner, these models lack the specificity necessary to fully define the process for onboarding.

C. Software Engineering Onboarding Models

Onboarding frameworks developed for Software Engineering teams build on these general principles by providing useful insights into documentation practices, mentorship, and knowledge transfer challenges [6-7]. While these insights are valuable, they do not provide a comprehensive model suitable for Security Engineering because they do not account for the privileged access, risk sensitivity, and governance requirements inherent in these roles.

D. Existing Organizational Attempts

Stakeholder feedback during initial interviews indicates that many organizations have made partial efforts to formalize onboarding for Security Engineering roles, but these attempts are typically fragmented, inconsistently applied, and overly dependent on individual initiative. Common artifacts, such as access checklists, group membership trackers, and informal documentation repositories, tend to focus on administrative tasks rather than providing a comprehensive, role-specific onboarding pathway. These materials rarely address elements such as access sequencing, mentorship, operational readiness, or cross-team integration. As a result, new hires often encounter unclear expectations, difficulty locating essential information, delays in access provisioning, and inconsistent exposure to subject matter experts. These patterns indicate that while organizations recognize the importance of onboarding, existing approaches lack the structure, governance, and security-specific considerations necessary to support early effectiveness.

The existing standards, organizational models, and prior attempts reviewed in this section illustrate that while the expectations for Security Engineering roles are well defined, the processes used to onboard individuals into those roles remain unstructured. This gap underscores the need for a systematic examination of how onboarding is currently practiced and where consistent challenges emerge across teams and organizations. To address this need, the next section outlines the methodological approach used to document existing practices, gather stakeholder perspectives, and derive the themes that informed the development of the proposed onboarding framework.

III. Methodology

This project used a mixed-method, design-oriented approach to identify onboarding gaps within Security Engineering teams and translate those insights into a structured onboarding framework. The methodological intent was to understand current onboarding practices, surface recurring challenges across stakeholder groups, and use those observations to inform the design of a practical, role-specific solution that will be discussed later. A mixed-method approach was selected because the problem involves both qualitative feedback and evaluation of existing processes, and a design-oriented investigation allows the findings to directly shape the proposed framework presented later. The methodological process consisted of two phases – baseline analysis and stakeholder feedback analysis.

A. Baseline Analysis

The first phase established an understanding of existing onboarding practices by reviewing organizational artifacts and process materials including onboarding checklists, access provisioning workflows, documentation repositories, and informal guidance used by Security Engineering teams. These artifacts were selected to capture the structural components of onboarding and to identify what information, processes, and support mechanisms were already in place.

Artifacts were catalogued and examined for structure, consistency, and role specificity. The review focused on how onboarding tasks were sequenced, how access was requested and approved, how documentation was organized, and what expectations were communicated to new hires. Materials that were not related directly to Security Engineering onboarding were excluded. This phase provided the procedural baseline against which stakeholder feedback could be compared. Examples of captured artifacts have been sanitized and include in Appendix A: Existing Onboarding Artifacts.

B. Stakeholder Feedback Collection

The second phase gathered qualitative data from individuals involved in the Security Engineering onboarding lifecycle. Semi-structured interview prompts were distributed asynchronously to Security Engineering managers, experienced engineers, and recent joiners. It was chosen to not schedule direct interviews due to the difficulty in scheduling because Security Engineers work constraints as well as to provide the ability to allow time to response thoughtfully to each question. Participants were selected to represent multiple organizations, varying levels of experience, and different roles in the onboarding process. The selection of participants reflects my personal network of contacts in the Security Engineering and closely related fields.

The interview prompts focused on onboarding experiences, perceived barriers, expectations, access readiness, documentation quality, mentorship structures, and early task confidence. A link to the interview document sent to participants can be found in Appendix B: Stakeholder Interview.

Responses were collected in written form to reduce scheduling constraints and encourage detailed reflection. Only responses addressing onboarding-related experiences were included. As the participants were assured confidentiality in these interviews, they contain personally and organizationally identifiable information. For this reason, the results of the interviews are not made public. 52 individuals were sent the invitation to participate in the interview and I received feedback from 23 individuals across 11 organizations. All responses were transcribed and grouped by stakeholder role.

An initial set of themes was generated using AI-assisted clustering to surface recurring concepts and terminology [8]. These clusters were then manually reviewed to ensure accuracy, remove noise, and align concepts with participant intent. Concepts were then merged into broad categories where appropriate. This process ensured that the analysis remained grounded in stakeholder input rather than researcher assumptions.

Applying this methodology yielded several notable themes that illustrate the nature of the problems encountered by new hires during onboarding. The following section presents these findings in a clear and objective manner.

IV. Findings

The thematic analysis conducted on stakeholder interview produced a consistent set of onboarding challenges that directly affect time-to-effectiveness in Security Engineering roles. Derived directly from the methodological process described in the previous section, these results are presented objectively and without interpretation. Across organizations and experience levels, participants described similar difficulties that affected the new hire experience. The following subsections present these results grouped into six themes, each representing a recurring onboarding gap that emerged from the analysis.

A. Role Clarity & Expectations

Stakeholders reported that new hires often begin without a clear understanding of responsibilities, expected performance, or early task progression. Interview responses described uncertainty about what to prioritize in the first weeks and how success would be evaluated. Several participants noted that the absence of defined 30/60/90 day expectations contributed to uneven ramp up. This theme reflects a recurring gap in how early role clarity is communicated.

B. Privileged Access Management

Participants consistently identified access provisioning as a major barrier to early effectiveness. New hires frequently encountered delays obtaining required permissions, and access workflows were described as opaque or dependent on manual approvals. Participants noted that these delays prevented completion of basic tasks and sometimes led to reliance on others to execute actions on their behalf. This theme highlights a persistent challenge in aligning access readiness with early job demands.

C. Documentation & Knowledge Transfer

Stakeholders described documentation as fragmented, outdated, or difficult to locate across multiple systems. New hires reported relying heavily on verbal explanations or repeated outreach to subject-matter experts to fill gaps. Several participants emphasized that essential information was often stored informally or inconsistently maintained. This pattern reflects a broader dependence on tribal knowledge during onboarding.

D. Mentorship & Support Structures

Interview responses indicated wide variation in the structure and consistency of mentorship. Some new hires experienced regular check-ins and guided exposure to tasks, while others described ad hoc or minimal support. Participants noted that inconsistent mentorship placed additional burden on experienced engineers and contributed to uneven onboarding experiences. This theme captures the variability in how support is provided during early role development.

E. Operational Readiness

Stakeholders reported that new hires often lacked early exposure to the organization's systems, workflows, and operational environment. Participants described uncertainty when interacting with tools or making changes due to limited initial context. Several noted that this gap increased the likelihood of errors or hesitation during early tasks. This pattern reflects challenges in preparing new hires for safe and confident operational work.

F. Psychological Safety & Team Integration

Participants emphasized that early social integration and reassurance were important components of onboarding. New

hires described benefits from environments where questions were welcomed and escalation felt safe. Stakeholders also noted that early relationship-building helped reduce hesitation when navigating complex responsibilities. This theme highlights the social factors that shape early confidence and engagement.

The themes presented in this section highlight that the issues new hires encounter are not isolated or organization-specific, but represent core onboarding gaps that occur across Security Engineering. These findings form the foundation for the design of the proposed onboarding framework presented in the next section.

V. Proposed Solution

Each theme identified in the findings guided one or more components of the structured onboarding framework. The design process emphasized clarity, repeatability, and alignment with the operational realities of Security Engineering work.

The onboarding framework follows a structured, sequential flow that moves through the onboarding process from role clarity to full operational readiness. It begins by defining the target by establishing responsibilities, required competencies, key systems, and capability boundaries. This anchors all subsequent onboarding activities. Capability boundaries provide explicit, risk-aligned task governance that allows the Security Engineer to demonstrate competence in a safe way. This provides a way to build confidence without presenting an unnecessary amount of risk. The framework also covers a milestone-based progression, using 30/60/90-day checkpoints to make development expectations, clear and measurable.

With the expectations clearly defined, the framework shifts to foundational clarity. In this section, it covers tips for ensuring that the new hire has the access, training, and documentation required to operate safely. A major point in this section is defining and implementing a knowledge infrastructure. This documentation provides a base that all of the other components rely on.

Now that the new hire is able to operate in their role, the framework goes on to cover development of operational capabilities which is where the onboarding process transitions from orientation to execution. Topics in this section include developing proficiency with the IT service management system, how to handle incident, and how operational readiness is validated.

Since effective Security Engineering is not only reliant on technical capability, the next section covers organizational complexities. Topics in this section include creating a structured mentorship program, integrating into the Security team, building relationships with other teams in the organization, preparing for being on-call, developing the expectations for communication and professional judgement, and validating organizational integration.

Now that the new hire has been brought into the organization properly, the framework covers how performance is measured. This is through a combination of validating milestones and operational metrics have been reached and checking that the new hire understands their role clearly. This section also includes guidance on what to do if new hires fail to meet or greatly exceed expectations set for them. Feedback from this section is also used to feed into onboarding process improvement.

As a final section, it was important to cover keeping the onboarding framework up to date with changes in the organization. Feedback from all other sections can be used to update and improve the process and make it fit the individual organization better. This section also includes setting up a review process and framework steward so that significant changes to the environment can be accounted for before they affect a new hire going through the process.

Decisions about structure, sequencing, and scope were guided by the need to create a model that could be applied across organizations with varying maturity levels while remaining specific enough to support Security Engineering roles.

Useable artifacts were developed and included in the framework to address gaps surfaced in the baseline analysis and stakeholder feedback collection phases. These include a role definition template, a Traffic Light Model template, milestone-based progression examples, knowledge infrastructure recommendations, and a policy-to-practice template.

The framework was refined iteratively using stakeholder feedback to ensure internal coherence and logical flow. Components were evaluated for completeness, alignment with identified patterns, and practical applicability. This framework is a living document; therefore, it is expected that changes will be made as feedback is received from usage. A change tracking table was added at the end of the document to reflect any updates. The final framework at 35 pages in length represents the synthesized output of the baseline analysis and stakeholder feedback coupled with personal experience and industry best practices.

A link to the onboarding framework is included for review in Appendix C: Onboarding Framework. Populated examples of the templates are included in the appendix of the onboarding framework to assist organizations with how the template should be filled in. The following subsections cover the high-level artifacts included in the framework, which findings they address, and how they do so.

A. Role Definition Template

The role definition template shown in Figure 1 addresses the findings of Role Clarity & Expectations and Privileged Access Management themes by providing a standard structure for communicating the expectations and requirements of the role to the new hire. The template includes sections for the most important information to convey. A mission summary defines where the role fits into the organization as a whole. Responsibilities provides a bulleted list of the high-level expectations the role is tasked with achieving. Key systems and competencies lists tools and topics that the new hire should be familiar with or seek further training in to function at the required contribution level. The final section contains a list of the security required to perform the job duties of the role. This list provides an easy reference to determine whether all permissions have been established early on to avoid bottlenecks from missing access.

Roles and Responsibilities

Title	Position Name	Department	Security Engineering
Mission Summary			
A brief description of what the job position entails including contributions the new hire will be expected to make to the team. The language used to fill in this template should always be clear and concise.			
Responsibilities			
• **A bullet list of the responsibilities the new hire will have**			
Key Systems & Competencies			
• **A bullet list of the requirements the new hire must meet to fulfil the needs of this position**			
Required Additional Security			
• **A bullet list of the additional security that people in this role must have to do their job**			

Figure 1 - Roles and Responsibilities Template

B. The Traffic Light Model

The Traffic Light Model shown in Figure 2 addresses findings of Privileged Access Management, Mentorship & Support Structures, Operational Readiness, and Psychological Safety & Team Integration themes by establishing explicit capability boundaries. It achieves this by categorizing tasks into groups based on whether the tasks can be performed independently (green), require supervision or review (yellow), or remain restricted until specific competencies or approvals are met (red). The template starts by describing the meaning of the color and the characteristics of tasks that should be placed in each bucket. It then provides example tasks for each color. This is where the organization would place actual tasks they feel match that color. The expectations section includes a list of themes that define what management is looking for from the new hire in work of that tier. The final section includes metrics that define how the new hire is assessed on tasks and what is necessary to advance to tasks of the next tier.

Color	Green	Yellow	Red
Meaning	Safe for early ownership and meant to build confidence and momentum	Require supervision or peer review but build competence in higher-risk activities	Restricted until demonstrated competence or milestone completion to protect critical systems and organizational risk posture
Characteristics	Low production impact Reversible actions Predefined runbooks Limited blast radius	Moderate impact Partial system exposure Change implementation risk Requires contextual judgement	High blast radius Irreversible or difficult rollback Regulatory implications High visibility impact
Examples	Processing standard access requests Updating documentation Triage of low-severity tickets Monitoring dashboard review Executing predefined scripts	Implementing configuration changes with review Reviewing complex access justifications Drafting change requests Handling medium-severity incidents Presenting findings to team	Privileged IAM policy changes Production firewall rule modification Major cloud architecture changes Incident commander role Approval authority for security exceptions
Expectations	Independent execution Clear documentation Daily queue engagement Escalation when unsure	Peer review before execution Shadow-to-lead progression Mentor sign-off required	Explicit authorization Milestone completion (90-day Review) Formal role transition On-call readiness validation
Measurable Indicators & Advancement Criteria	Number of independently resolved tickets Documentation quality review score SLA adherence	Demonstrated consistency in Green Zone Successful completion of 30/60-day milestones Manager approval	

Figure 2 - Traffic Light Model Template

C. Milestone Progression

The milestone progression addresses findings of Role Clarity & Expectations, Documentation & Knowledge Transfer, Mentorship & Support Structures, Operational Readiness, and Psychological Safety & Team Integration themes by providing a structured, time-bound path that makes new hire development repeatable, predictable, and less risky. It gives a list of expectations for the new hire to work toward across 30, 60, and 90-day milestones. The framework provides recommendations on creating this milestone list as well as giving examples of suggested goals for each check step. Figure 3 shows a list of example goals for the 30-day milestone. This list reduces ambiguity, enables early detection of gaps, and ensures that new hires advance toward operational readiness in a manner consistent with management's expectations instead of relying on guesswork.

Example Milestones:

30-day:

- All required accounts and access provisioned and confirmed
- Demonstrates successful navigation and understanding of core systems (ITSM, SIEM, etc.)
- Successfully completes 1-2 guided tasks with mentor support
- Show comfort navigating team communication channels
- Observe one change review meeting and one change readiness meeting

Figure 3 - 30-Day Milestone Examples

D. Knowledge Infrastructure

The knowledge infrastructure addresses the findings of Role Clarity & Expectations, Documentation & Knowledge Transfer, and Operational Readiness themes by providing a centralized repository with an organized structure for documentation, which provides reference material and operational guidance necessary for the new hire to become effective. It provides guidance on standardizing where and how documentation is stored, which reduces reliance on informal knowledge transfer and individual memory. Figure 4 provides a list of recommended documentation types to include in the knowledge infrastructure. By making this critical information discoverable, current, and consistently structured, the knowledge infrastructure enables faster onboarding and supports repeatable workflows. This ensures that new hires have confidence when undertaking actions as they progress in responsibilities.

Area	Description
Architecture	Diagrams and system documentation that explains how everything connects, such as flows, dependencies, and components an engineer will interact with
Knowledge Base	Includes frequently asked questions (FAQs), quick-reference guides (QRGs), and end-user documentation that help engineers understand how systems work
Policy-to-Practice	Contains short, practical breakdowns of key policies and how they apply to day-to-day Security Engineering work
Roles	Stores the role definitions and Traffic Light Models for each role in the team to provide clarity and set expectations around task readiness
Runbooks and Playbooks	Provides step-by-step guides for the operational processes an engineer will encounter

Figure 4 - Knowledge Infrastructure Documentation Types

E. Policy-to-Practice Template

The policy-to-practice template shown in Figure 5 addresses the findings of Documentation & Knowledge Transfer, Operational Readiness, and Psychological Safety & Team Integration by translating security policies that are often filled with complicated wording into clear, operational guidance for Security Engineers. It breaks each policy into

Policy Name	The formal name of the policy
Plain-Language Summary	Describe what the policy governs in straightforward, practical terms. Focus on operational impact, not legal language.
Practical Impact on Security Engineer	Explain how this policy affect day-to-day engineering work, including decisions, tasks, and workflows.
Where the Policy Applies	List the systems, processes, environments, or data types where this policy is relevant.
Engineer Responsibilities	Outline what the engineer must do to comply with the policy. Include required actions, prohibited actions, and judgement calls.
Controls That Enforce the Policy	Identify the administrative or technical controls tied to this policy, such as approval workflows, logging, access restrictions, or monitoring rules.
Escalation Triggers	Define the conditions that require escalation to a manager, GRC, Legal, or other team. Keep this section unambiguous.
Comon Failure Points	Note common mistakes or misunderstandings engineers encounter when applying this policy
Related Documentation	Link to runbooks, playbooks, standards, diagrams, or procedures that support this policy in practice.

Figure 5 - Policy-to-Practice Template

While these components are presented separately, the framework is designed to function as a cohesive system. Together, these components create a repeatable onboarding process that reduces ambiguity, accelerates. The framework was designed to be adaptable, allowing organizations to tailor role definitions, capability boundaries, and knowledge infrastructures to reflect their architecture, risk profile, and security program maturity. In the next section, I will discuss the evaluation of this framework and how well it achieved its goal of improving onboarding time-to-effectiveness.

VI. Evaluation & Results

This section describes the evaluation conducted to assess the effectiveness of the proposed solution and present the resulting data. The evaluation focused on how well the solution addressed the needs identified in the Findings section and how well the solution itself is implemented. A mixed-method approach was used, using a structured survey to collect both quantitative and qualitative feedback from participants. This section is organized into three parts: evaluation design, evaluation procedures, and results.

A. Design

Comparative measurement of onboarding outcomes requires observing at least one full onboarding cycle per group, which for most organizations spans a minimum of 90 days. Because this semester timeframe did not allow for a full before-and-after assessment and because the framework itself had to be developed prior to evaluation, the study relied on proxy indicators rather than direct measurement of time-to-effectiveness. The evaluation used role clarity, operational readiness, and onboarding efficiency which have been shown in research to contribute to improvements in time-to-effectiveness [3-5]. Each proxy indicator was then broken down into 4 subfactor which contributed to that indicator.

B. Procedures

The evaluation was performed and the results collected using an online survey created using Microsoft Forms [8]. This platform allowed for a variety of question types and the ability to provide the necessary guidance to the user at each stage of the evaluation.

A section was provided for each proxy indicator and a general framework assessment where a Likert-scale assessment was provided for each subfactor with a description of what that subfactor means. This provided the quantitative assessment for each section. Two qualitative questions were also asked for each section which allowed freeform feedback from the participants. This feedback was used to assess the components that had the biggest impact or could be improved.

I sent the survey to the 23 participants from the initial interview. The survey period was 3 weeks. During this time I received responses from 18 of the participants which represented 8 organizations. A link to the full evaluation survey can be found in Appendix D: Evaluation Survey.

C. Results

This section presents the results of the evaluation survey. Below you will see the results of the Likert-scale questions for each indicator of improved time-to-effectiveness during onboarding. The overall averages for responses to these

questions were [SA: 58.34%; A: 34.17%; N: 7.24%; D: 0%; SD: 0.28%]. The full results of the survey are available in Appendix C: Evaluation Survey. In the following subsections, I will cover the results as they apply to each indicator and to the assessment of the framework overall because these taken separately is more valuable than the combined averages. Each subsection includes the average percentage of responses for selection option and a summary of what the qualitative feedback indicated for each indicator.

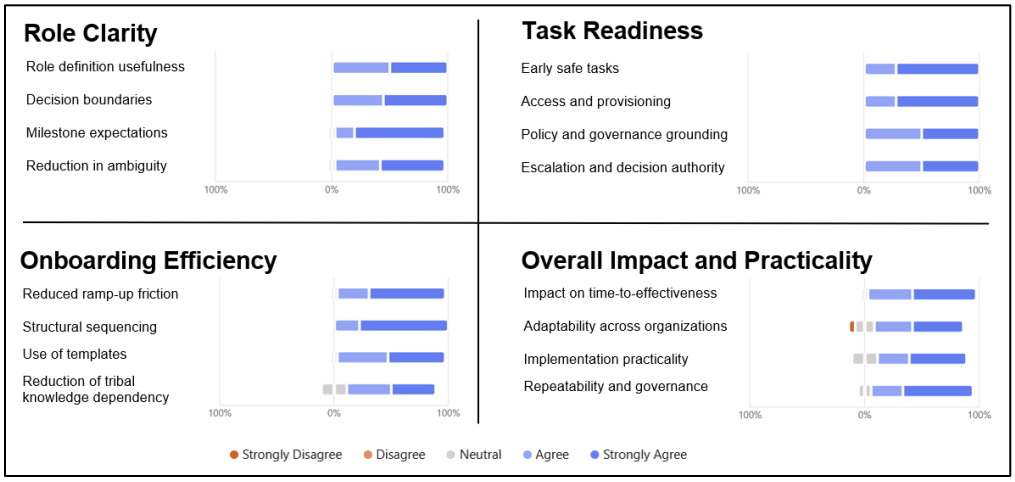

Figure 1 - Evaluation Survey Results (Adapted from Microsoft Forms Output) [8]

1) Role Clarity

[SA: 59.75%; A: 37.5%; N: 2.8%; D: 0%; SD: 0%]

Participants strongly agreed that the framework improves early role clarity. The role definition template, Traffic Light Model, and 30/60/90-day milestones were consistently identified as the most impactful components. Participants described those elements as providing structure, clear expectations, and baseline operational understanding. The primary limitation noted was the need for role-specific tailoring, especially for specialized domains such as cloud security or detection engineering. Several responses highlighted the need for clearer boundaries with adjacent teams and guidance beyond the initial 90 days.

2) Task Readiness

[SA: 61.1%; A: 38.9%; N: 0%; D: 0%; SD: 0%]

The participants indicated that the Traffic Light Model was the strongest contributor to task readiness, frequently describing it as confidence-building, empowering, and essential for safe early work. Participants also emphasized that access and provisioning guidance addresses a common organization bottleneck, and that policy/governance grounding and escalation authority help new hires make decisions aligned with audit and risk exceptions. Recommendations for enhancement included scenario-based exercises, role-specific examples, and formal sign-off checklists to validate readiness for higher risk tasks.

3) Onboarding Efficiency

[SA: 58.35%; A: 33.33%; N: 8.35%; D: 0%; SD: 0%]

Participants agreed that the framework would reduce onboarding friction and create a more predictable experience. The most impactful elements were structured sequencing, access provisioning, templates, and knowledge infrastructure. These components were mentioned as reducing ad hoc work and accelerating time-to-effectiveness; however, participants noted that some inefficiencies, such as cross-team dependencies, documentation quality, and mentor availability, are organizational issues that the framework cannot fully eliminate.

4) Overall Practicality and Impact

[SA: 52.78%; A: 31.95%; N: 13.9%; D: 0%; SD: 1.4%]

Participants generally agreed that the framework is adaptable, realistic to implement, and likely to reduce time-to-effectiveness. Several responses noted that only minor organization-specific tweaks would be required. The negative feedback centered around the initial lift required for organizations with low documentation maturity, with one response noting that the framework assumes a mature process with documentation. Despite this, the consensus was that the framework provides a strong, repeatable structure that meaningfully improves onboarding outcomes across all evaluation criteria.

These results establishes the empirical evidence that the framework would positively contribute to improving Security Engineering onboarding time-to-effectiveness. In the next section, I will discuss the interpretation of these findings and evaluate the framework's strengths, limitations, and implications.

VII. Discussion

This section interprets the results from the previous Evaluation & Results section and explains what they indicate about the effectiveness of the proposed onboarding framework. While the previous section presented data, the focus in this section is on interpreting what that data means in relation to the

original problem statement and the Findings identified earlier in this paper. First, I will cover the overall performance of the framework based on the evaluation results. Afterwards, I will move on to examine its strengths, limitations, and what the results imply for organizations that implement the framework and for future work.

A. Overall Performance

The consistent clustering of Strongly Agree and Agree responses indicate that the framework would meaningfully improve these early indicators of improved time-to-effectiveness. These patterns suggest that the framework performs well across the dimensions most critical in the early stage of a Security Engineer's onboarding. The results also show that the framework directly addresses the themes identified in the Findings section, particularly around reducing ambiguity, structuring early tasks, and improving the sequencing of onboarding activities.

B. Strengths

The evaluation results also highlight several strengths in the design and structure of the proposed onboarding framework. First, the framework provides a clear end-to-end sequence that reduces the mental load typically placed on new hires by distributing information and tasks in a manageable and timely way. Participants consistently highlighted the role definition, Traffic Light Model, and 30/60/90-day milestones as the most valuable components. These elements reduce ambiguity, establish shared expectations, and give new hires a clear understanding of their responsibilities and progression. This structured progression supports early confidence and reduces reliance on ad hoc knowledge transfer. Second, the framework's modular structure allows organizations to adapt the process to their specific environments while maintaining a consistent baseline of expectations and outcomes. This balance between standardization and flexibility allows teams with different technical stacks, maturity levels, or operational constraints to implement the framework without losing consistency or comparability. Finally, the framework's emphasis on clear role definition, early task structure, and team integration supports predictable onboarding experiences and reduces reliance on ad hoc knowledge transfer, which participants noted as a common pain point in prior onboarding experiences.

C. Limitations

Despite the strengths of the proposed onboarding framework, a few of limitations constrain the interpretation and general applicability of the evaluation results. First, as mentioned in the Methodology section, the freeform text provided in the stakeholder interviews contained personally and organizationally identifiable information that could not be disclosed in this report. A different format or coding of the interview could have provided a method to prevent this to increase transparency of the methodology. Second, as discussed in the evaluation design section I was unable to perform a before-and-after evaluation of the time-to-effectiveness of the onboarding process to directly measure the impact of the proposed framework on time-to-effectiveness. Basing evaluation on proxies provides an indirect way to measure impact. While this evaluation captures perceived improvement in early onboarding conditions, it does not empirically measure the performance outcome of the framework. Third, the perspectives gathered were limited by the size and composition of the participant pool. The evaluation reflects the views of 18 participants from 8 organizations which represent a variety of stakeholder roles. Even with the variation of perspectives, the evaluation may not capture the full range of challenges faced by teams with different maturity levels, technical stacks, or onboarding models. Broader participation would strengthen the validity of the results and increase the adaptability inherent in the framework.

These limitations do not diminish the value of the evaluation but instead clarify the boundaries within which the findings should be interpreted.

D. Implications

The evaluation results indicate that the onboarding framework can materially improve role clarity, operational readiness, and onboarding efficiency which should lead to improved time-to-effectiveness without the need for new tooling or organizational restructuring. By formalizing role expectations, capability boundaries, and milestone progression, the framework reduces reliance on informal knowledge transfer.

Organizations implementing this framework can strengthen their onboarding outcomes by treating it as an operational process rather than an administrative activity. The framework's artifacts provide a mechanism for aligning new hire development with the architecture and workflows specific to the organization. This alignment enables earlier detection of capability gaps, more consistent knowledge transfer, and clearer paths to effective individual contribution.

The evaluation also suggests broader implications for workforce development and organizational maturity in Security Engineering teams. A standardized onboarding framework creates a foundation that is repeatable and creates better team integration. It positions teams to scale more effectively by reducing variability introduced by differences in onboarding cycles. While the evaluation was limited in scope, the perceived impact suggests that structured onboarding can serve as a low-cost, high-leverage factor for improving security posture and engineering productivity.

This discussion highlights that the onboarding framework provides meaningful operational value and where further refinement is needed. In the next section, I will summarize the contributions this project has made to solving the stated problem.

VIII. CONCLUSION

This project set out to address a persistent challenge in Security Engineering: the lack of structured, repeatable onboarding processes that reliably accelerate time-to-effectiveness. Through interviews and thematic analysis, the work identified findings in the form of themes which defined

common pain points experienced during onboarding. These findings guided the development of a structured onboarding framework designed specifically for Security Engineering teams, integrating role definitions, capability boundaries, milestone-based progression, and knowledge infrastructure guidance.

The evaluation results demonstrate that the framework meaningfully addresses the core problems surfaced in the research. Participants consistently agreed that it reduced ambiguity, strengthens early decision-making, and provides a predictable, sequenced onboarding experience. The Traffic Light Model, role definition elements, and 30/60/90-day milestones were especially impactful, offering a practical and risk-aware way to guide new hires through increasingly complex responsibilities. At the same time, the evaluation highlighted important considerations for implementation, including the need for role-specific tailoring, documentation ownership, and cross-functional coordination.

This project contributes a practical, adaptable solution that organizations can use to improve the reliability and effectiveness of Security Engineering onboarding. It fills the current gap shown in the Background & Related Work section and addresses the Findings with targeted artifacts and supportive documentation. While the framework cannot fully resolve organizational issues such as cross-team dependencies or documentation maturity, it provides a strong foundation for organizations seeking to mature their onboarding processes and reduce time-to-effectiveness. Together, these efforts can help Security Engineering teams to build a more repeatable and scalable onboarding experience.

IX. Future Work

Along with the direct implementation of the onboarding framework, the project revealed other areas where continued work on the topic could act to extend on the foundation this project has built. These areas include refinement of the framework contents, broader validation by more organizations, and the inclusion of more templates to ease implementation.

First, direct implementation of the framework into a Security Engineering team would allow an opportunity to observe real-world usage, gather operational feedback, and assess its impact on time-to-effectiveness in practice. Second, refinement of the framework's contents could better clarify expectations, define decision points, and incorporate additional guidance where early users identify ambiguity. Third, further evaluation across a wider range of organizations would allow for comparison of onboarding needs, maturity levels, and security operating models. Finally, expanding the set of templates and supporting material would reduce implementation overhead and make the framework easier to adopt at scale.

Collectively, these directions would extend the contribution of this project by providing practical testing of the framework, strengthening the framework's clarity, assessing its applicability across a wider range of organizational environments, and expanding the practical resources that support its implementation. They outline a path for continued refinement and maturation of the onboarding framework, ensuring it can evolve alongside the needs of Security Engineering teams and deliver increasing value over time.

X. References

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XI. Appendices

A. Appendix: Existing Onboarding Artifacts

Examples of existing onboarding artifacts are available for review at: https://github.com/dldishner/Security-Engineering-Onboarding-Framework/blob/main/Existing%20Onboarding%20Artifacts.pdf

B. Appendix: Stakeholder Interview

The initial stakeholder interview sent out to participants is available for review at: https://github.com/dldishner/Security-Engineering-Onboarding-Framework/blob/main/Initial%20Interview.pdf

As the participants were assured their feedback would remain confidential and only the aggregation of the data would be released, they provided feedback which contained information which could be used to identify them or their organizations. As a result, I am unable to make the results of the stakeholder interviews available for review as part of this paper. If you would like additional clarity about anything relating to these results, please reach out to me directly.

C. Appendix: Onboarding Framework

The full onboarding framework including the added example populated templates appendix is available for review at: https://github.com/dldishner/Security-Engineering-Onboarding-Framework/blob/main/Onboarding%20Framework.pdf

D. Appendix: Evaluation Survey

The evaluation survey used in assessing the proposed solution is available for review at: https://github.com/dldishner/Security-Engineering-Onboarding-Framework/blob/main/Evaluation%20Survey.pdf

The results of the survey are available for review at: https://github.com/dldishner/Security-Engineering-Onboarding-Framework/blob/main/Evaluation%20Survey%20Results.xlsx

XII. AI USAGE DISCLAIMER

In the process of working on this project, I have used (or tried unsuccessfully to use in some cases) Microsoft Copilot [9], ChatGPT [10], and Grammarly [11] as supportive tools. The portions of the project where these tools were used:

- Project ideation to develop ideas from personal strengths and interests and ensure alignment with the requirements of the course [10]
- Structure and flow refinement on my project documents to ensure they match the expected academic structure [9]
- Evaluation methodology suggestions to best capture the viability of the solution while avoiding bias and leading questions [9]
- An overall sounding board for the direction my project is going and whether I may have forgotten to include something [9]
- Grammar analysis of writing to avoid commonly made mistakes in punctuation, continuity, or clarity [9, 11]

Copilot [9] was used to do an initial clustering of themes from the free text input of the stakeholder interview feedback. This clustering was reviewed manually to ensure that the assessment was accurate and correctly identified the intention of the stakeholder in their response.

All analysis, interpretations, and final decisions presented in this report were vetted with my own judgement and feedback received from individuals who contributed responses to surveys and interviews. AI-generated suggestions were reviewed, validated, and edited to ensure accuracy, clarity, and alignment with project requirements.